

HARIKRISHNAN H

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Education

College of Engineering Trivandrum

Master of Technology in Artificial Intelligence

2025 – Present

Trivandrum, Kerala

College of Engineering Perumon

Bachelor of Technology in Computer Science

2021 – 2025

Kollam, Kerala

Core Competencies

- Web Development (Full)
- Database Management
- Flutter App Development
- API development
- Model Engineering
- Full Stack Integration
- Automation workflows
- Scripting

Research Interests

- Adversarial AI & ML
- AI Security & Privacy
- Deepfake Detection
- Generative AI & LLMS
- AI Red Teaming
- Full Stack AI Integration
- AI in Agriculture

Experience

Freelance Machine Learning Engineer

Smart Parking Detection System

Nov 2025

[View Code](#)

- Developed a high-precision Parking Spot Classifier using a custom Convolutional Neural Network (CNN) in **PyTorch**.
- Built a 3-stage CNN feature extractor using Conv2d and MaxPooling to detect complex spatial patterns.
- Designed a robust fully connected classification head with varying Dropout rates (0.5, 0.3) to prevent overfitting.
- Achieved **100% classification accuracy** on the test dataset, validating the model's reliability for real-time detection.

Freelance AI Engineer

Smart Stress Detection System

Oct 2025

[View Code](#)

- Engineered a multi-modal AI system combining lifestyle data analysis and real-time facial emotion recognition using Flask and OpenCV.
- Developed a **Random Forest/XGBoost** model, achieving an R^2 score of **0.91** on synthetic data.
- Architected a custom 5-layer **CNN** in **PyTorch** trained on the FER2013 dataset to classify 7 distinct emotions.
- Designed a weighted fusion algorithm to refine stress metrics using real-time emotional feedback..

Freelance Frontend Developer

Pencil Labs (Frontend Engineering)

Oct 2025

[View Site](#)

- Architected a high-performance, type-safe web interface using **React**, **TypeScript**, and **Vite**.
- Developed a scalable component library and responsive UI patterns to ensure consistent UX across devices.
- Optimized application bundle size and load times, leveraging Vite's modern build tooling for superior performance.

Freelance Frontend Developer

Front End Developer Intern

Sept 2025

[View Site](#)

- Developed a modern, responsive platform for Personal Strategic Planning™ using **React**, **TypeScript**, and **Vite**.
- Implemented a scalable and accessible design system utilizing **shadcn-ui** and **Tailwind CSS** for consistent UI/UX.
- Optimized production builds and component architecture to ensure high performance and cross-device compatibility.

Freelance GenAI Developer

AI Voice Interviewer (Generative AI & LLM)

Mar 2025

[View Code](#)

- Built a voice-interactive interview platform using a locally hosted **DeepSeek-R1 (1.5B)** model via **Ollama**.
- Enabled real-time voice interaction by integrating STT/TTS pipelines with **React** and **Flask**.
- Implemented RAG-style logic to parse PDF resumes and dynamically generate context-aware technical questions.
- Optimized prompt engineering strategies to extract structured feedback and scoring from lightweight local models.

Freelance Developer

Workflow Automation Script (Google Apps Script)

Feb 2025

[View Code](#)

- Developed a serverless automation solution using **Google Apps Script** to streamline data reporting workflows.
- Integrated Google Forms and Gmail APIs to trigger instant, custom email notifications.
- Eliminated manual monitoring by automating the parsing and distribution of response data.

Projects

Crop Yield Prediction <i>Python, Scikit-learn, Pandas</i>	Nov 2025
- Conducted a comparative analysis between Linear Regression and Decision Tree models to predict agricultural yields. - Demonstrated the superiority of Decision Trees in handling non-linear dependencies, achieving an R^2 score of 0.98 significantly outperforming Linear Regression (0.75). - Reduced Root Mean Squared Error (RMSE) from 4.21 to 1.20, validating the model's suitability for sustainable agriculture applications.	
To-Do App <i>Dart, Flutter</i>	Feb 2025
- Developed a cross-platform mobile application for task management using the Flutter framework. - Implemented efficient local storage mechanisms to persist user data and task states across sessions.	
Snake Game Infinite <i>Godot Engine, GDScript</i>	Aug 2024
- Developed a multi-platform game (Web, Android, PC) featuring dynamic scoreboards and random generation. - Designed a user-friendly interface with responsive character controls and adaptive gameplay mechanics.	
3D Portfolio Website <i>React, Three.js, Tailwind CSS</i>	July 2024
- Led the creation of a dynamic 3D portfolio website implementing complex 3D features. - Customized design and content using React and Three.js for an immersive user experience. - Optimized performance and responsiveness using GSAP and Tailwind CSS.	

Technical Skills

Languages: Python, Java, C, HTML/CSS, JavaScript, SQL, Dart, GDScript
AI & Machine Learning: TensorFlow, Keras, Pandas, Matplotlib, NumPy, cv2
Web Development: React, Node.js, Flask, HTML5, CSS, Tailwind, Javascript, Three.js, Vite, Next.js
Tools & Platforms: VS Code, Git, Github, Docker, Postman, Linux/Unix, Firebase, MySQL, Android Studio, Godot

Leadership / Achievements

Hackathons & Events	2023 – 2026
<i>Winner & Organizer</i>	<i>India</i>
1st Prize Winner (Student Category) India AI Impact Buildathon 2026: Engineered a multi-lingual AI voice fraud detection API, achieving 95%+ accuracy against high-fidelity deepfakes with an ultra-low 4ms latency. - First Prize Winner, Hack with CEP: Led team to victory in a 24-hour innovation hackathon. - Head Coordinator, INCEPTO: Organized a national-level hackathon, overseeing everything from planning to logistics for the 24-hour event. - Coordinator, MathVenture: coordinated "MathVenture," a tech event centered on mathematical problem-solving. Oversaw event planning, operations, and participant engagement.	

Leadership & Community

Community Lead & Mentor <i>TinkerHub, LeetCode</i>
- Founded and facilitated a LeetCode community, mentoring peers in Data Structures and Algorithms. - Served as a Volunteer Mentor for multiple TinkerHub hackathons and events. - Organized and conducted a Web Development workshop, teaching fundamentals to junior undergraduates.
Event Participant <i>StackUp, Saturday Night Hack</i>
- Collaboratively architected a dynamic, responsive web platform using Next.js during the StackUp event. - Engaged in rapid prototyping and development during bi-weekly "Saturday Night Hack" sessions.

Certifications & Workshops

UAS/Drone Technologies Boot Camp: Completed a one-week offline program on Allied Technologies for Drones facilitated by NIELIT Calicut and organized by IEEE Kerala Section.
Cyber Security & Bug Bounty: Mastered vulnerability assessment techniques and strategies for mitigating online security risks and safeguarding digital assets.
Industrial Automation (PLC): Designed automated systems using Programmable Logic Controllers to optimize material and hardware-based industrial processes.
Certifications: Coursera: Fundamentals of C, OOP in Java, Introduction to Python, and Frontend Development. Hackerrank: Problem Solving (Basic) Certificate.